MODULE 05 Random Variables Probability Distribution

P of an event
P of success P(getting a head)
P of failure P(not getting Head) 1-P(getting a h
Conditional Probability P of A given probability B
Independent/Dependent events

" Probability tells us how likely something is. But when we perform a random experiment, how can we represent its outcome numerically?"

Coin Toss
Possible Outcomes = {H,T} = S
Head = 0
Tail = 1

X = Random variable

Therefore :

P(H) X(H) = 0
P(T) X(T) = 1

A random variable does not mean the variable itself is randomly changing without reason.
It means the numerical value assigned to an outcome depends on the result of a random Experiment.

Random Variable:
A random variable is a numerical values asssigned to the outcome of a random experiment

Example:
Roll a die
S = {1,2,3,4,5,6}

Define:
X= number appearing on the die

Outcome	Random Variable X
1	1
2	2
3	3
4	4
5	5
6	6

MODULE 05 Random Variables Probability Distribution

X: S--> R

A random variable is a function that maps each outcome of a random experiment to a real number

S= Sample Space

X = Random Variable

R = Set of Real Numbers

Example : Number of Head

Experiment: Toss a coin twice.

S = {HH,HT,TH,TT}

Define:

X = Number Of heads

Outcome	No. Of Heads	X
HH	2	2
HT	1	1
TH	1	1
TT	0	0

X ∈ {0,1,2}

The experiment produces:

HH,HT,TH,TT

But the random Variable produces:

2,1,1,0

Q. Is a random variable the same as an outcome?

No

An outcome is the result of the random experiment

A random variable assigns a numerical value to that outcome

Business Example E-Commerce

Suppose an e-commerce company receives orders every hour.

Define:

X = Number of orders received in one hour

Possible Values: 0,1,2,3,4,.....

Hour	Orders
9:00 AM	12
10:00 AM	18
11:00 AM	15
12:00 PM	23
1:00 PM	19

It is telling the order count from 8 AM to 9AM

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Random Experiment: Observe one hour of orders.

X= Number of orders in an hour

Can X = 18.5 orders?

No

Why?-- The orders can be counted there can't be any partial order.

Business Usecase:

1. Retail

X = Number of customers entering the store

2. E-Commerce

X = Number of orders received

3. Banking

X = Number of Transactions

4. Customer Support

X = Number of Calls received

5. Manufacturing

X = Number of defective products.

DISCRETE RANDOM VARIABLE

A discrete random variable can take countable numerical values

Examples: no of customers, defects and other things that can be counted

Discrete = Count

Example 1: Dice

	X = number on the die
Possible Values	{1,2,3,4,5,6}

Example 2: Number of Defects

A factory produces 100 products.

Suppose	X = Number of defective products
Possible Values	{0,1,2,3,.....100}
You can't say there is X = 3.5	defective products

Example 3: Customer Purchase

A customer visits an online store.

X =number of the products purchased

Possibile {0,1,2,3,4,.....}

Again countable--> discrete

NOTE: **If you can answer the question using " HOW MANY?", it is often a strong idication that the variable is discrete.**

MODULE 05 Random Variables Probability Distribution

Transaction	No. of items
T-001	2
T-002	3
T-003	4
T-004	5
T-005	2
T-006	6

No of items is discrete bec

Could an order contain 2.7
No
Number of transaction = N
COUNT() counts the obser
Variable

CONTINUOUS RANDOM VARIABLES

A continuous random variable can take any value withing a given interval.

Continous = Measure

Example: -- Height

X = height of a person

possible values could be
170cm
170.5cm
170.52cm
170.527cm
170.5278cm and so on
There can be infinitely many possibilities for value between two numbers

Example -- Customer Waiting time

X = customer waiting time

Possible
2 Minutes
2.5 Minutes
2.53 Minutes
2.537 Minutes etc
Unlike number of customers Time is measure.

Example: Product Weight

A manufacturing company produces packets

X = weight of packet.

Possible:
500 g
500.2 g
500.25 g
500.257 g etc
Therefore It is continous

500

10 packets --> 150.2g Net
150.200010
150.199990
150.200010
150.200000

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Business Usecase:

1. Manufacturing

Product Weight --> continuous

2. Healthcare

Patient temperature --> continuous

3. Logistics

Delivery time -->continuous

4. Finance

Stock return percentage--> percentage

5. Customer Service

Waiting Time--> continuous

DISCRETE VS CONTINUOUS

Feature	Discrete	Continuous
Basic idea	Count	Measure
Values	Countable	Any value in interval
Example	Number of Customers	Customer Waiting time
Example	Number of defects	Prodcut Weight
Example	Number of calls	Temperature
Decimal Values	No menaingful	Common
Typical question?	How Many	How Much? / How Long? /How heavy?

What is a random variable?

A numerical function that assigns a numerical value to each outcome of a random Experiment

What is discrete random variable?

A random variable that takes countable numerical values

What is continuous random variable?

A random variable that can take any value withing a given interval

Number of Cutomer--> Discrete

Customer Waiting Time--> Continuous

Number of defective products?--> Discrete

Product weight?--> Continuous.

MODULE 05 Random Variables Probability Distribution

P(A|B)

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Discrete Random Variable

6

vation, it does not calculate the random